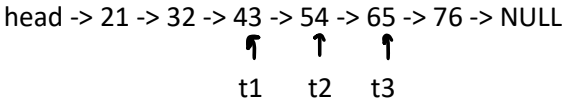
Quiz # 5 (Data Structures) CLO2 **BS DS Fall 2025 Afternoon**

Name:_____ **Roll number:_____**
Time Allowed: 15 min **Total Marks: 30**

Q1. For a singly linked list, find the difference between the sum of values at odd positions and the sum of values at even positions.

```
template <class T>
T LinkedList<T>::differenceOddEvenPositions()
{
   Node<T>* temp = head;
   T oddSum = 0, evenSum = 0;
   int position = 1;
   while (temp != NULL)
   {
      if (position % 2 == 1)
         oddSum += temp->info;
      else
         evenSum += temp->info;
      temp = temp->next;
      position++;
   }
   return oddSum - evenSum;
}
```

Q2. Consider the following linked list. The pointers t1, t2, and t3 point to specific nodes. Perform all operations in order, redraw the resulting list, and write the output of the print statement.



t3->next = t1;
t1->next = head;
head->next = t2;
t2->next = NULL;
 	
for (t1 = head; t1 != NULL; t1 = t1->next)
  cout << t1->data << ' ';
	Output:
	____**21 54**_____
	
	New List:
	
	____ **head → 21 → 54 → NULL**____

Q3: Consider the following two function part of Linear singly link list. What is the output of the given main function.

```
template <class T>
void LSLL<T>::F3()
{
   head = F3Helper(head);
}

template <class T>
Node<T>* LSLL<T>::F3Helper(Node<T>* node)
{
   if (node == NULL || node->next == NULL)
      return node;

   Node<T>* second = node->next;
   node->next = F3Helper(second->next);
   second->next = node;

   return second;
}

int main()
{

   LSLL<int> list;
   list.insertAtHead(23);
   list.insertAtTail(5);
   list.insertAtHead(4);
   list.insertAtTail(2);
   list.insertAtHead(6);
   list.insertAtTail(8);
```

```
    list.insertAtHead(9);

    list.F3 ();
    list.display();
```

```
return 0;

}
```

6 9 23 4 2 5 8